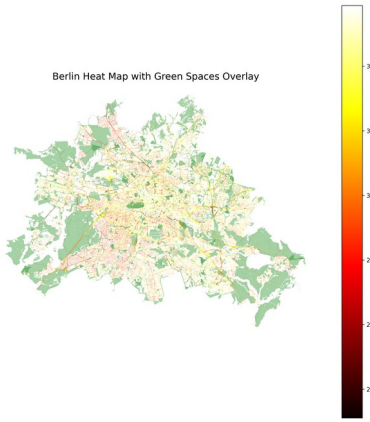
Berlin Heat Risk & Cooling Priority Explorer

Geospatial urban-climate decision-support project that maps Berlin heat patterns, compares green vs non-green areas, and prioritizes cooling interventions.

GIS | Urban climate | Decision-support | Streamlit app

Positioning

Not just a heat map — a planning-oriented tool for translating spatial heat patterns into cooling-priority decisions.



Heat map with green-space overlay



Cooling priority map

PROBLEM FRAMING

Why this project matters

Urban heat is unevenly distributed, but stakeholders need a workflow that turns spatial analysis into intervention logic.

Planning problem

- City-wide averages hide local heat differences.
- A hotspot map alone does not tell decision-makers where to intervene first.
- Green-space context helps explain where cooling may already exist.
- A portfolio-ready output should connect analysis, prioritization, and usability.

From data to action

- Use official Berlin surface-temperature data.
- Overlay green-space polygons from OpenStreetMap.
- Summarize differences by district.
- Flag non-green hot zones as cooling priorities.
- Package the outputs in a Streamlit dashboard.

DATA AND METHODS

Inputs and analytical workflow

The project combines climate, land-cover, and administrative data, then turns them into charts, maps, CSV outputs, and a dashboard.

Surface temperature

Berlin Climate Analysis 2022 surface-temperature layer

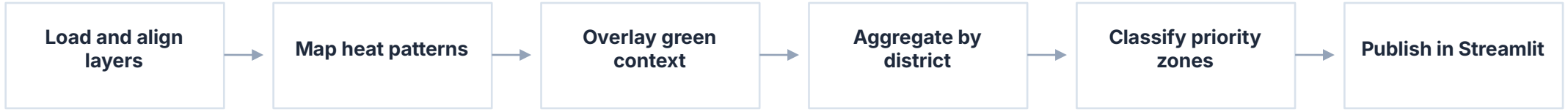
Green-space polygons

OpenStreetMap polygons to distinguish green from non-green context

District boundaries

Administrative units for district-level aggregation and summaries

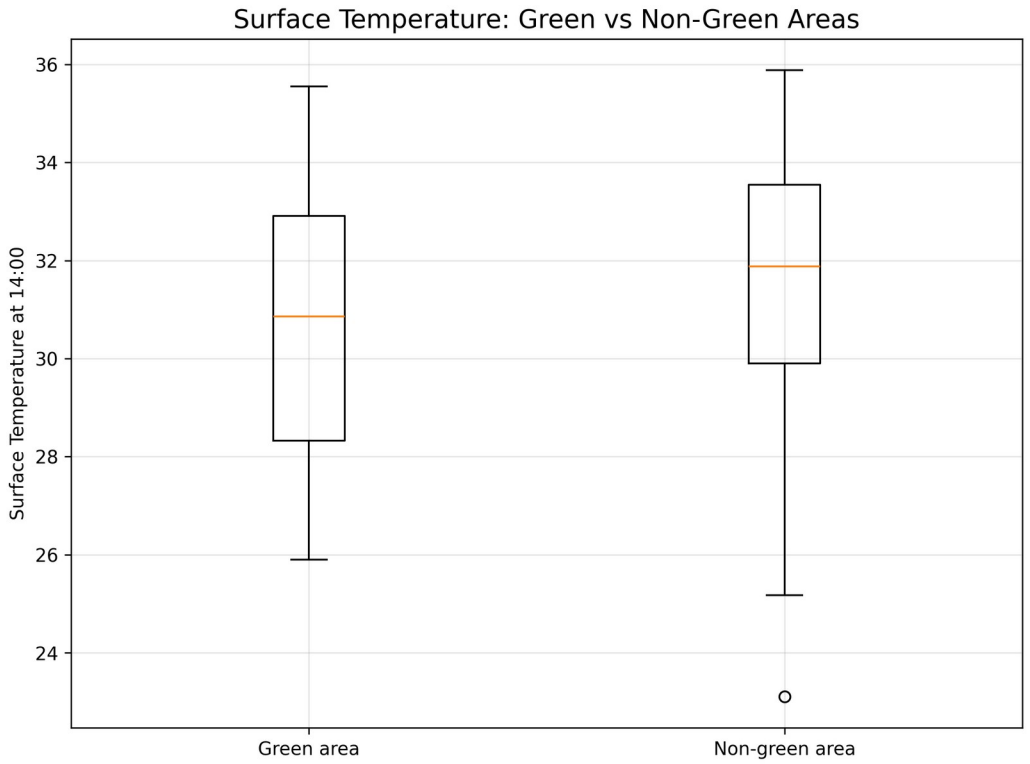
Method pipeline



KEY EVIDENCE

Green areas are cooler on average

This is the clearest quantitative result in the project and a strong bridge to urban greening and cooling interventions.



Green vs non-green temperature comparison

Takeaways

30.66° C Mean in green areas

31.68°C Mean in non-green areas

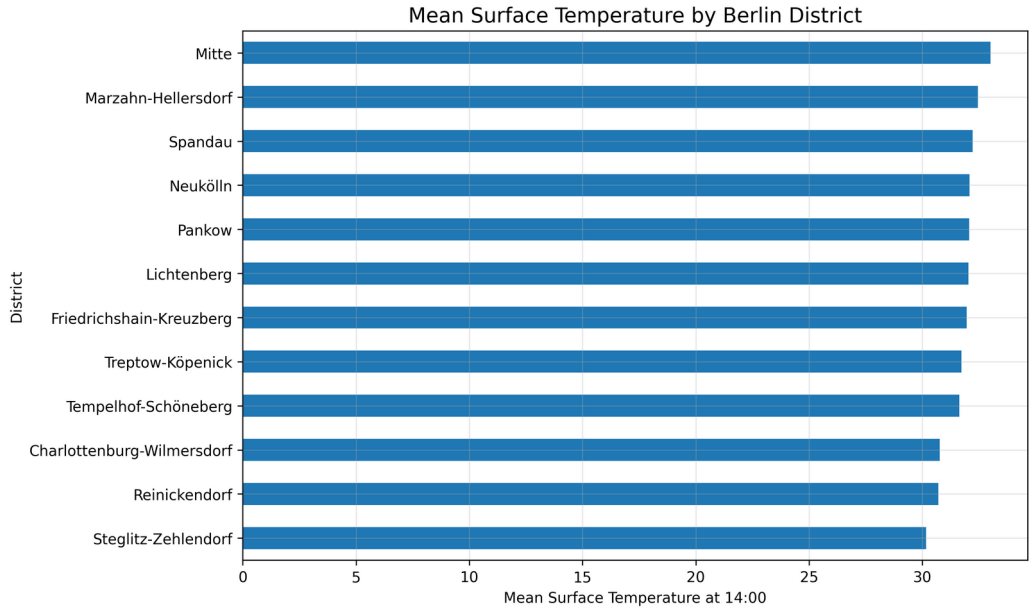
Green areas were 1.02°C cooler on average.

- Makes the comparison evidence-based rather than purely visual.
- Supports the relevance of urban greening, shade, and cooling strategies.
- Creates a strong headline result for the presentation.

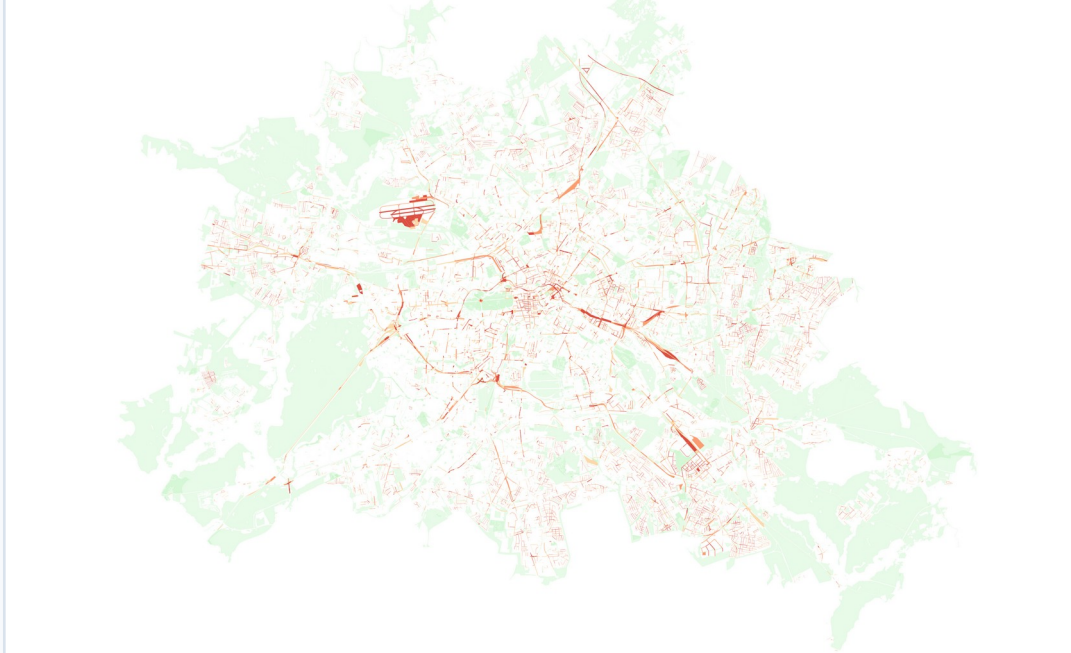
SPATIAL OUTPUTS

District variation and intervention priorities

The district chart adds planning readability, while the priority map turns heat patterns into a targeting workflow.



District mean temperature summary



Cooling priority map

District summaries improve interpretability; the priority layer makes the project more than descriptive mapping.

PRODUCT LAYER

Maps, outputs, and dashboard delivery

The project already produces static visual outputs plus a Streamlit app for exploration.

Outputs and app features

- Static outputs include maps, a boxplot, district summaries, and ranked priority-zone CSVs.
- The Streamlit app adds KPI cards, district and priority filters, an interactive cooling-priority map, tables, and a figures tab.
- Best next step: deploy the app publicly so reviewers can interact with it.

Current app status

- Works locally in Streamlit.
- Includes KPI cards, filters, map, tables, figures tab, weather context, and overview insights.
- Ready for public deployment after final polish.

PORTFOLIO VALUE

Why this project is stronger than a normal class submission

It combines spatial analysis, urban-climate framing, product thinking, and communication-ready outputs.

Geospatial analytics

- Spatial overlay and aggregation
- District-level summaries
- Priority-zone classification

Urban-climate framing

- Heat-risk interpretation
- Green vs non-green comparison
- Cooling intervention narrative

Product delivery

- Interactive dashboard
- Maps + CSV exports
- Portfolio-ready communication

Interview framing

Frame it as an urban-tech decision-support tool that turns geospatial analysis into actionable cooling-priority insight.

CLOSE

Limitations, next steps, and Q&A

Finish by showing maturity: what the project already does well, what it does not claim, and how you would extend it.

Current limitations

- Uses a specific surface-temperature layer, not live city temperatures.
- Surface temperature is not the same as direct human thermal comfort.
- Priority logic does not yet include vulnerability or sensitive infrastructure.

Best next upgrades

- Add demographic vulnerability indicators.
- Integrate schools, hospitals, and sensitive infrastructure.
- Include canopy cover, land use, or scenario-based intervention layers.
- Deploy the Streamlit dashboard publicly.

Q&A prep

- Why is this more than a heat map? Because it adds comparison, district aggregation, and a priority layer.
- What is the key result? Green areas were 1.02°C cooler on average.
- What would you add next? Vulnerability, canopy, infrastructure, and scenario-based layers.